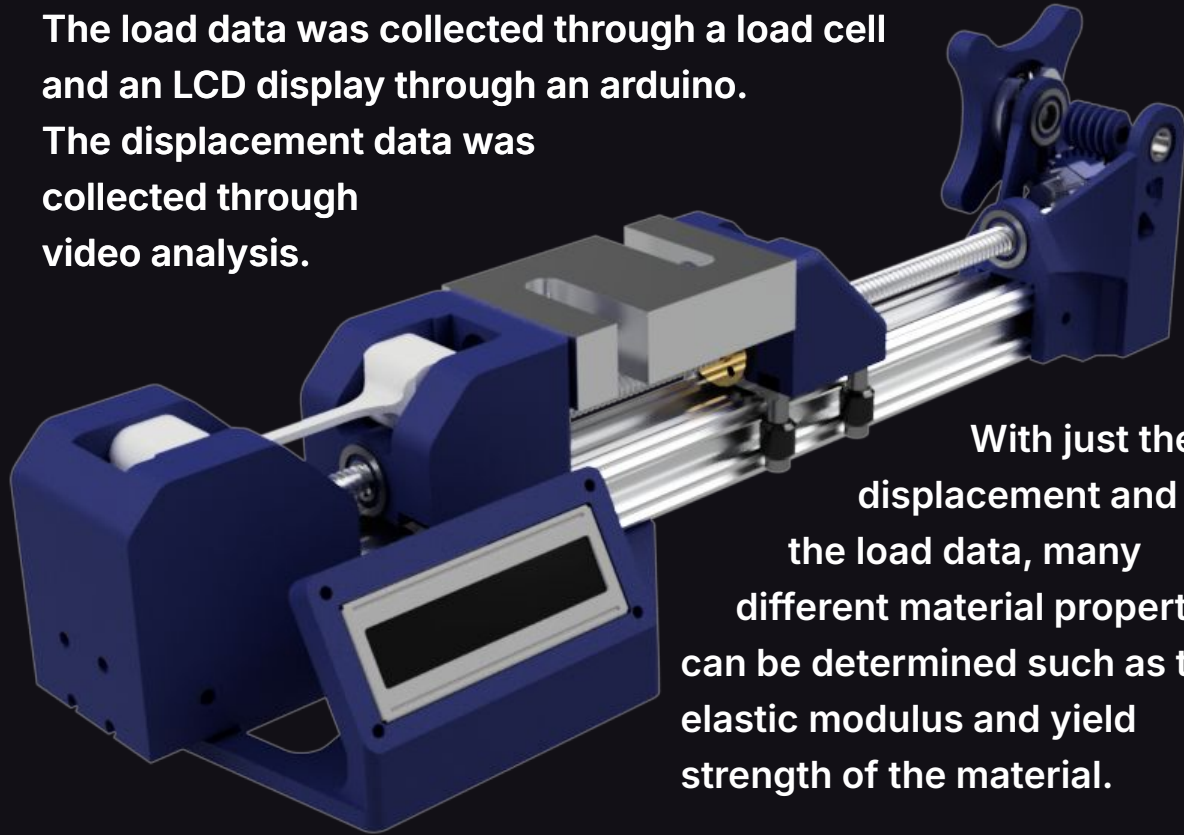


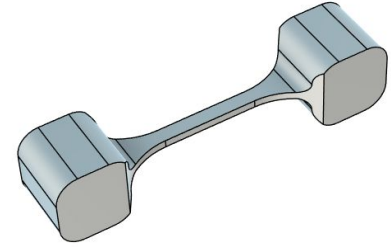
Design and Procedure

The load data was collected through a load cell and an LCD display through an arduino.

The displacement data was collected through video analysis.



With just the displacement and the load data, many different material properties can be determined such as the elastic modulus and yield strength of the material.

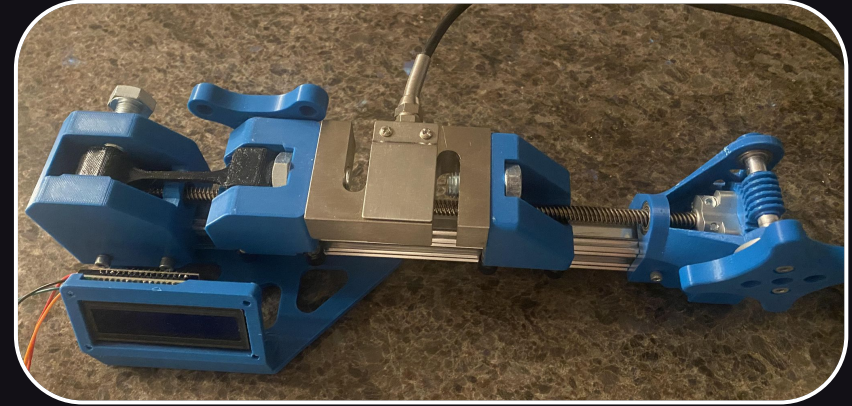


The load samples were designed in this dog bone shape and made from many different filaments.



Data Analysis

After obtaining data from the experiment, the stress vs strain graph can be fit to an approximate curve. Here is the process of taking a PLA sample from experiment to interpretation.



$$\sigma = \frac{F}{A} \quad \epsilon = \frac{\Delta L}{L}$$

The setup was far from ideal, so an approximate curve was made from many trials.

